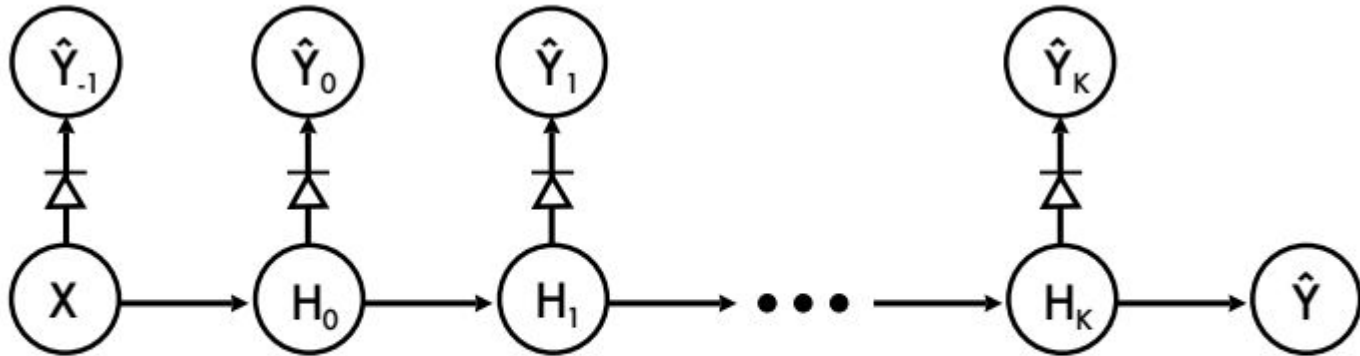


Can Sparse Autoencoders
disentangle superposed
features?

Mechanistic Interpretability

Representational Probing

(Alain & Bengio, 2016)

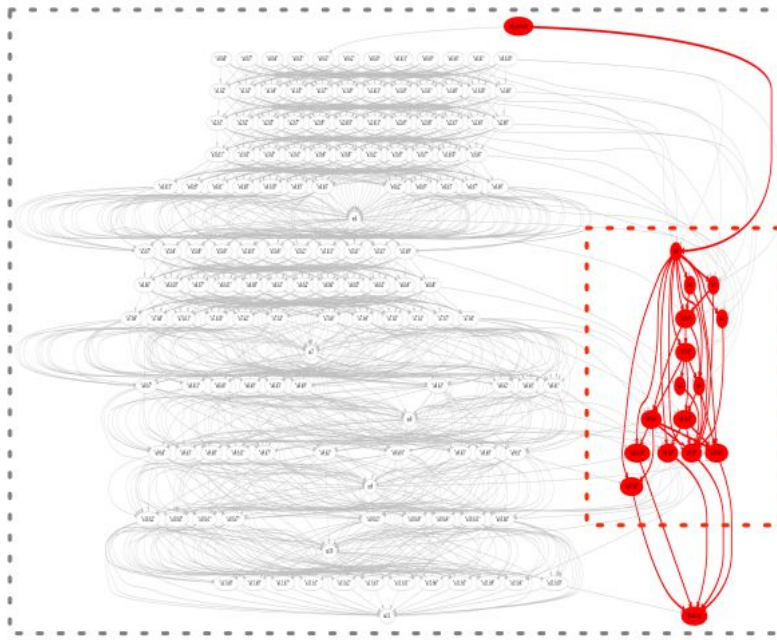


Mechanistic Interpretability

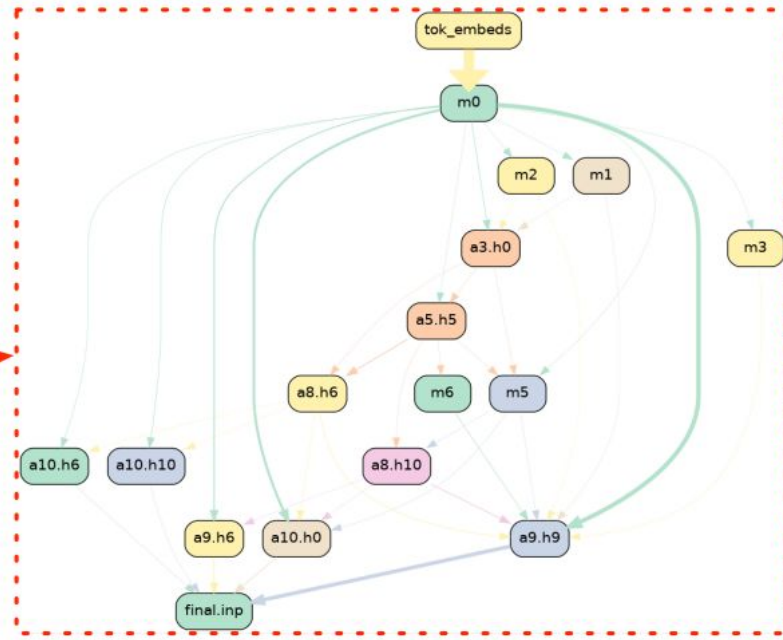
Circuits

(Conmy et al., 2023)

GPT-2 Small

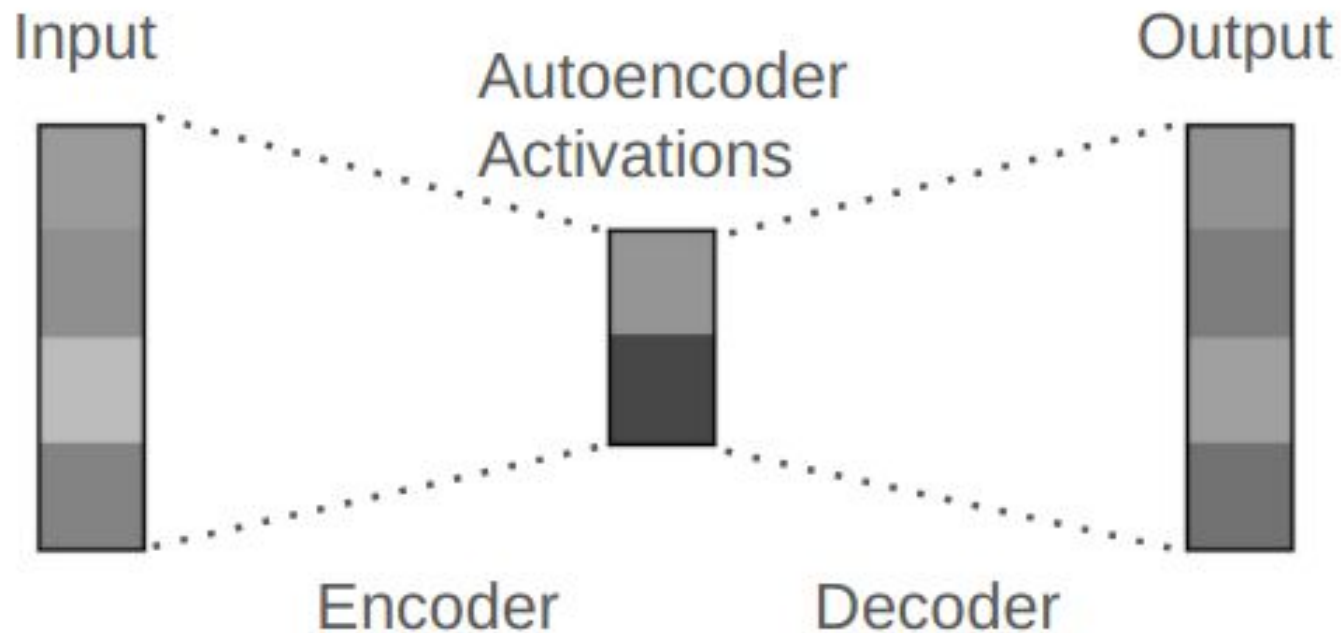


ACDC Circuit



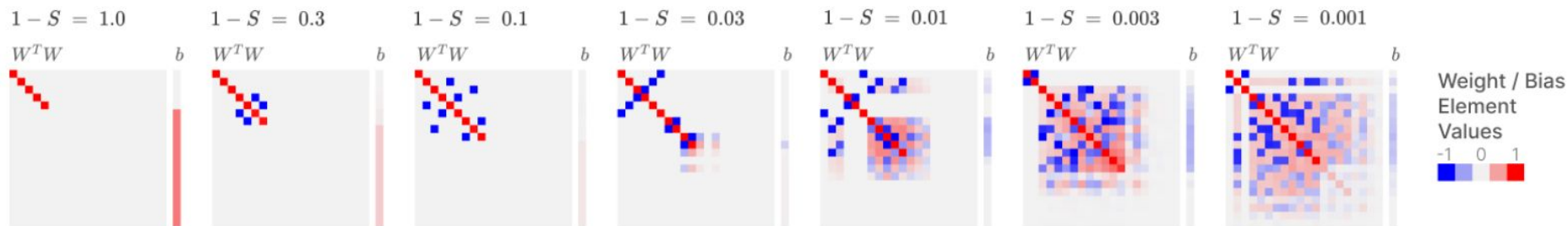
Superposition

$$x' = \text{ReLU}(W^T W x + b) \quad (\text{Elhage et al., 2022})$$



Superposition

ReLU Output Model

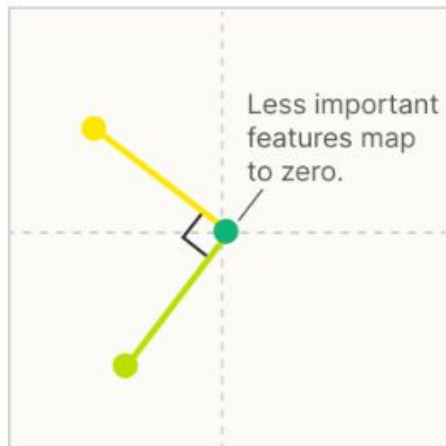


(Elhage et al., 2022)

Superposition

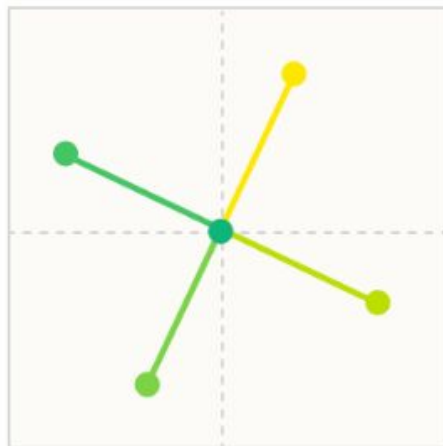
As Sparsity Increases, Models Use “Superposition” To Represent More Features Than Dimensions

Increasing Feature Sparsity →



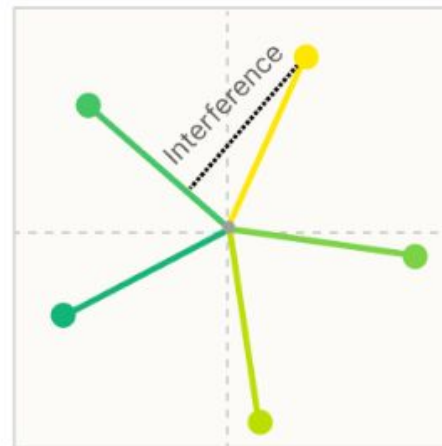
0% Sparsity

The two most important features are given **dedicated orthogonal dimensions**, while other features are **not embedded**.



80% Sparsity

The four most important features are represented as **antipodal pairs**. The least important features are **not embedded**.

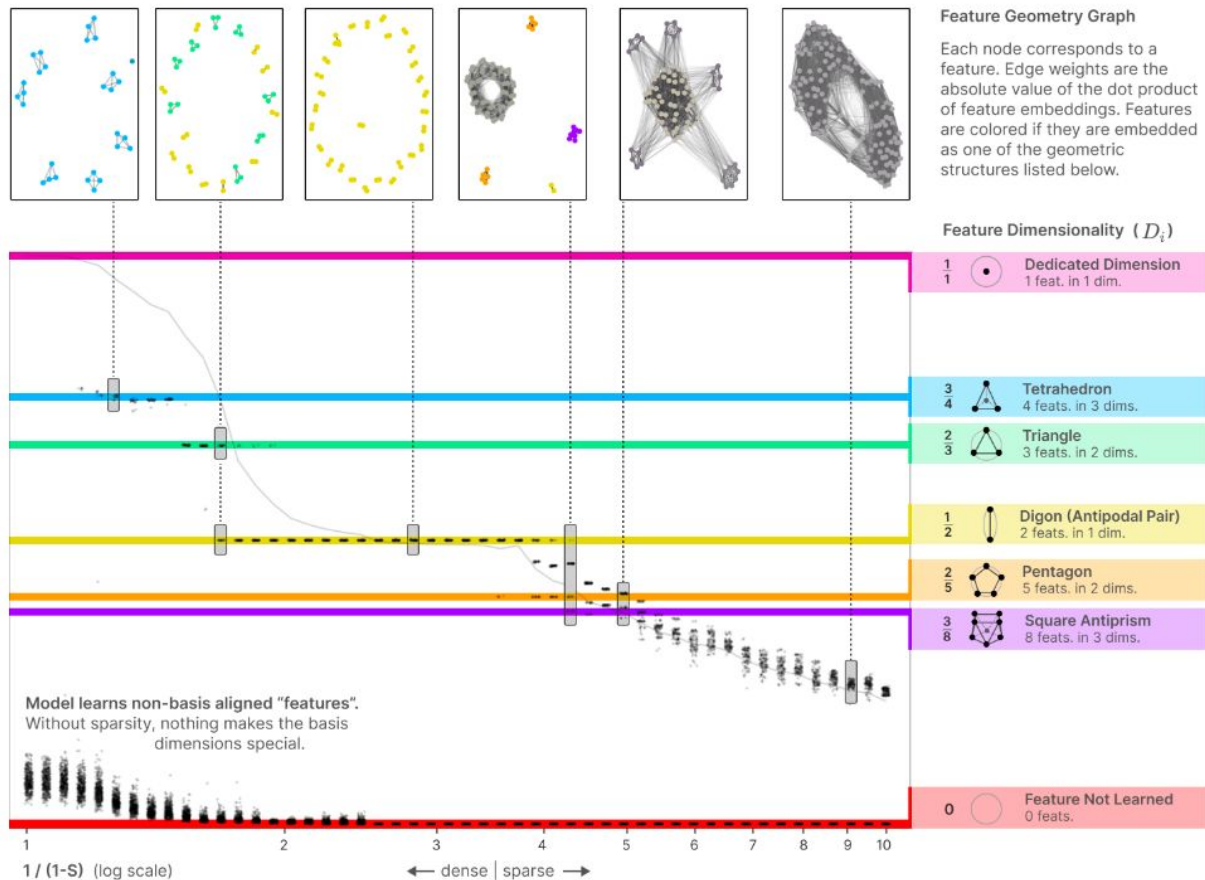


90% Sparsity

All five features are embedded **as a pentagon**, but there is now “positive interference.”

(Elhage et al., 2022)

Superposition

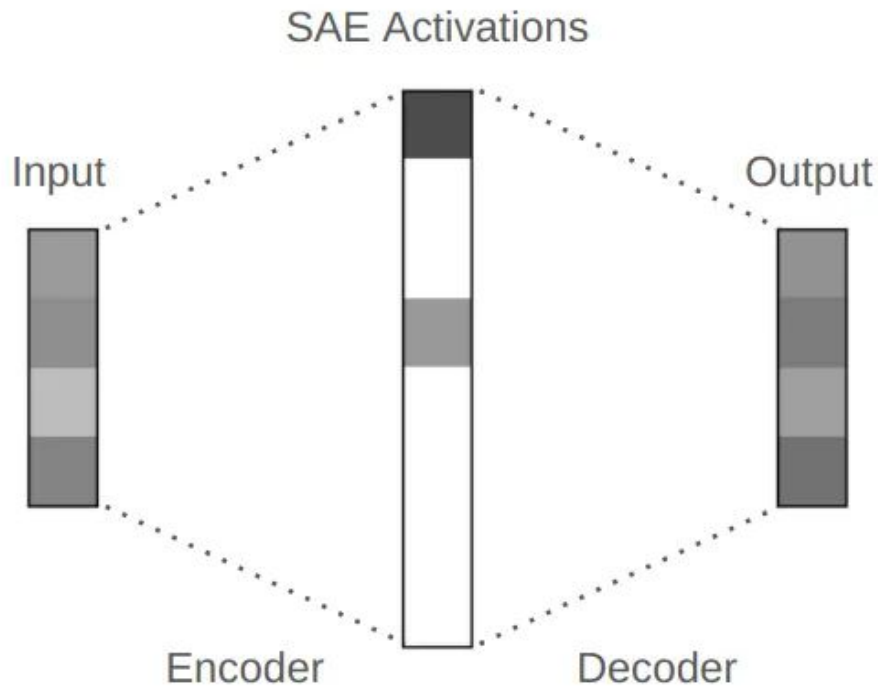


(Elhage et al., 2022)

Mechanistic Interpretability

Sparse Autoencoders

(Karvonen, 2024)



Mechanistic Interpretability

Sparse Autoencoders

#1332 Movies / TV / Fiction (esp. action, superhero) ?

AUTOINTERP. (SCORE = 0.219) ?

The neuron fires on references to comic books, superheroes, and related pop culture topics like superhero movies.

NEURON ALIGNMENT ?

Neuron	Value	% of L ₁
353	+0.36	3.2%
182	+0.26	2.4%
138	-0.23	2.1%

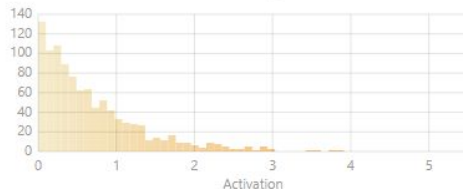
CORRELATED NEURONS ?

Neuron	Pearson Corr.	Cosine Sim.
#182	+0.14	+0.14
#353	+0.11	+0.11
#286	+0.09	+0.10

CORRELATED B FEATURES ?

Feature	Pearson Corr.	Cosine Sim.
#1007	+0.70	+0.70
#400	+0.10	+0.11
#3677	+0.10	+0.10

ACTIVATIONS (DENSITY = 0.8228%) ?

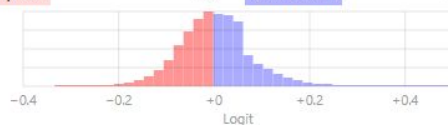


NEGATIVE LOGITS ?

gag	-0.33
Address	-0.33
orta	-0.32
ortic	-0.31
cester	-0.30
irrigation	-0.30
ort	-0.30
%	-0.30
{}	-0.29
Endpoint	-0.29

POSITIVE LOGITS ?

Batman	+0.50
thriller	+0.49
Marvel	+0.49
comics	+0.49
comic	+0.48
villain	+0.48
protagonist	+0.48
storyline	+0.48
films	+0.48
characters	+0.47



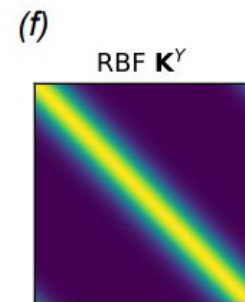
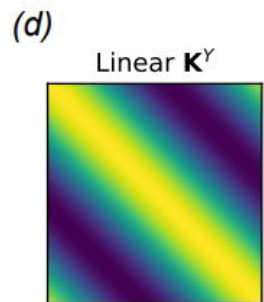
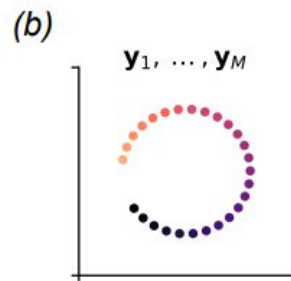
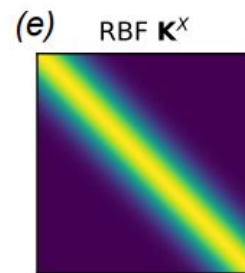
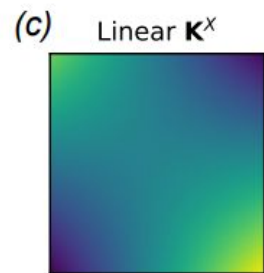
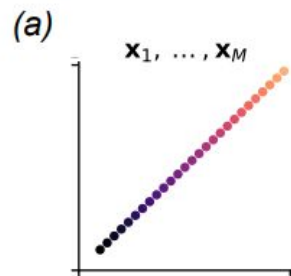
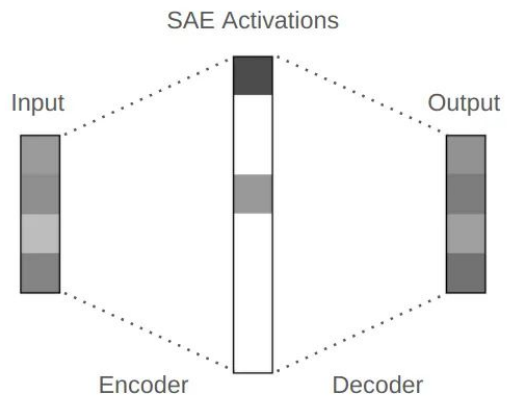
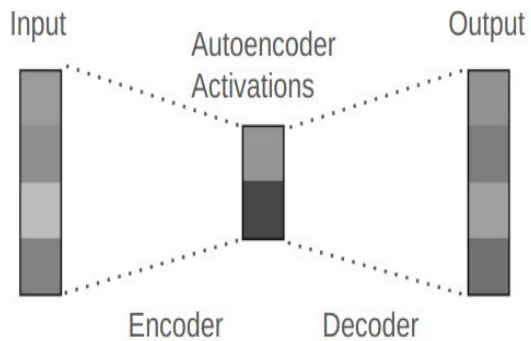
TOP ACTIVATIONS ?

TRAIN TOKEN MAX ACT = 5.200

who gains Superman-like abilities after being bit
also an unlikely paring of a duck named
the film deconstructing himself. Tastes
Spider-Man films featuring Toby McGuire
better than most DC animated movies, there are
ly accurate when dealing with the idea of why
require superhero-sized budgets, some film
Date With Diana, both he and Speedy
Superman, try to separate themselves from the Marvel
ism is quite jarring; we're
the fantasy genre-defining pedestal and effectively
lines DC has ever produced. And
in this film's cast). I just wish
in a thoroughly captivating fashion. Into Dark
things as the reanimated cat and the use
with a middle-aged Top Gun playing tough
comic book film, unlike movies like 'Man
over, unlikable characters, no real
jokes, charismatic characters, good hardcore
depth. Sure, CGI is fun, but

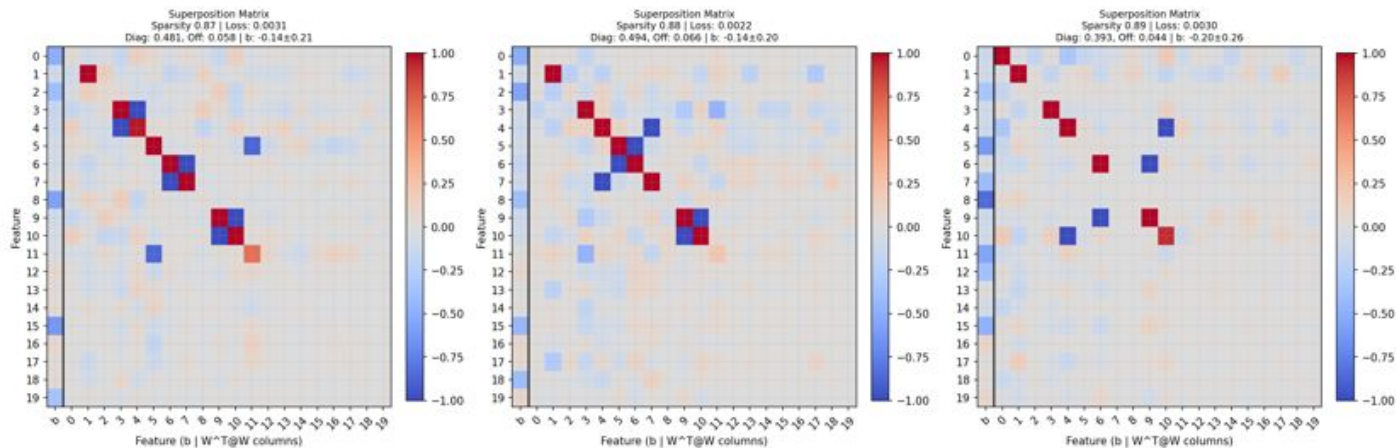
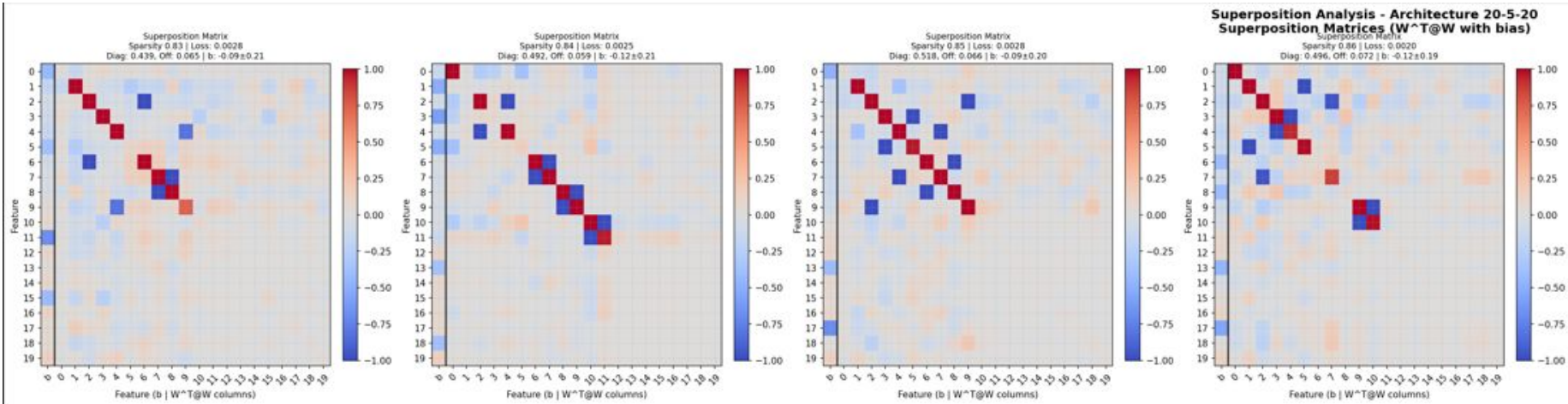
(Bricken et al., 2023)

Experiment

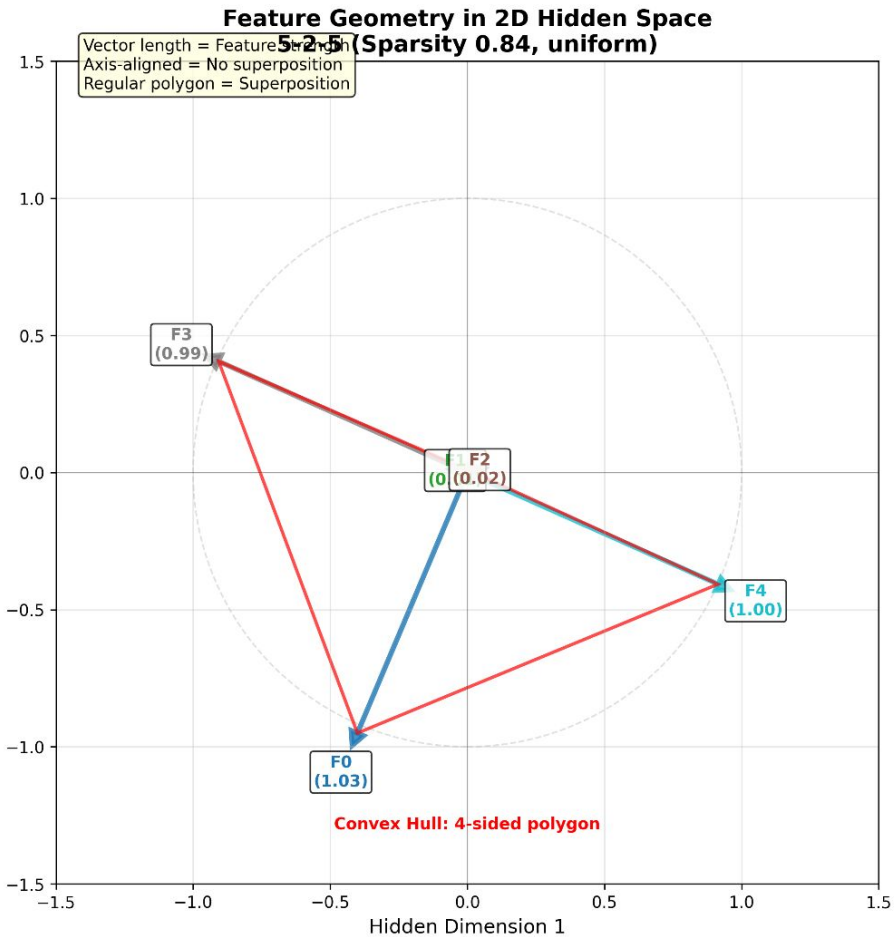
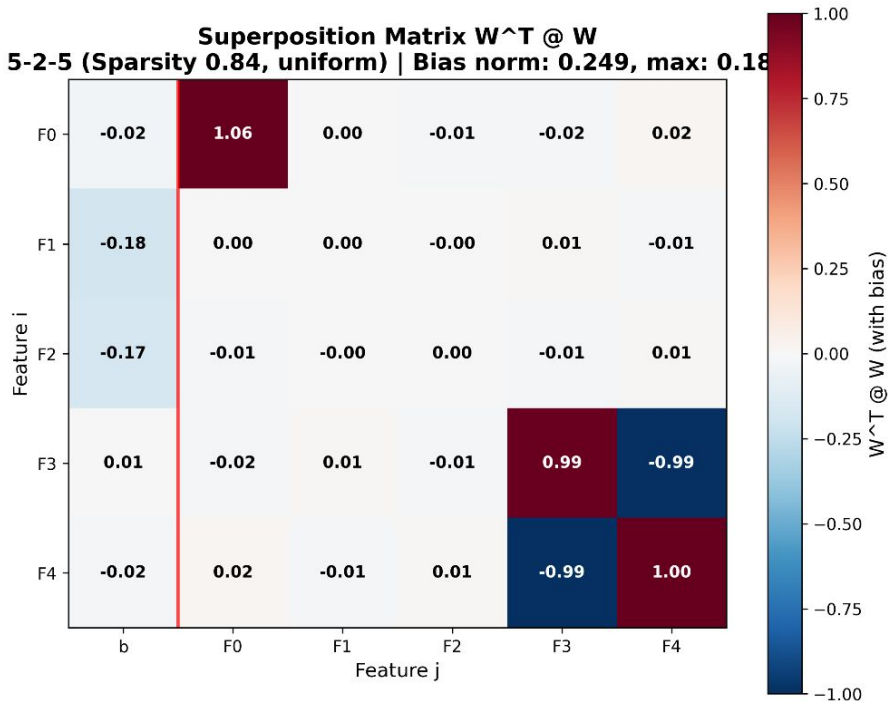


(Williams, 2024)

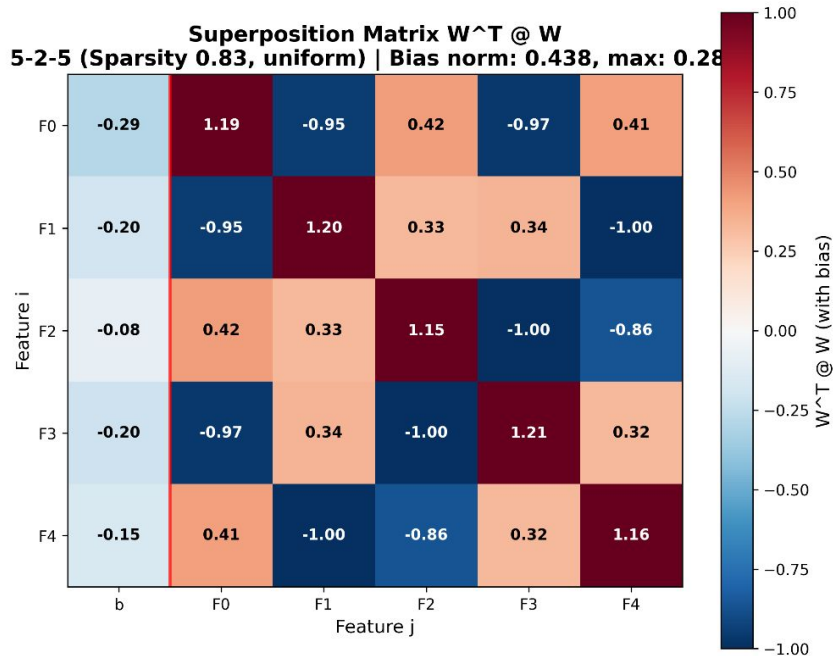
Results



Superposition Matrix $W^T @ W$
5-2-5 (Sparsity 0.84, uniform) | Bias norm: 0.249, max: 0.18

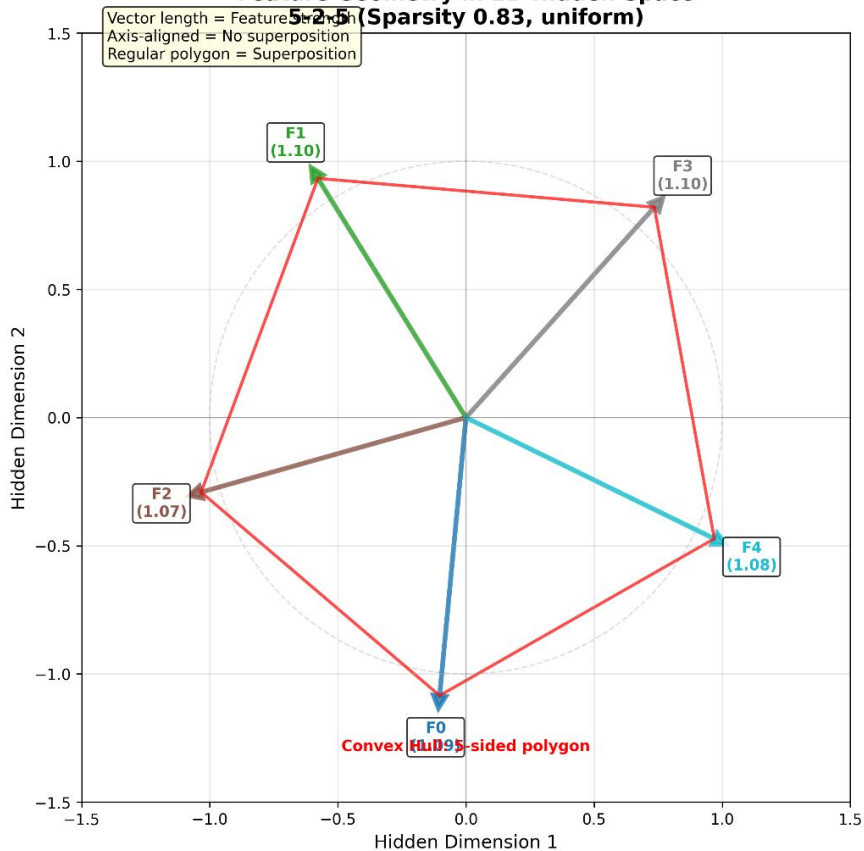


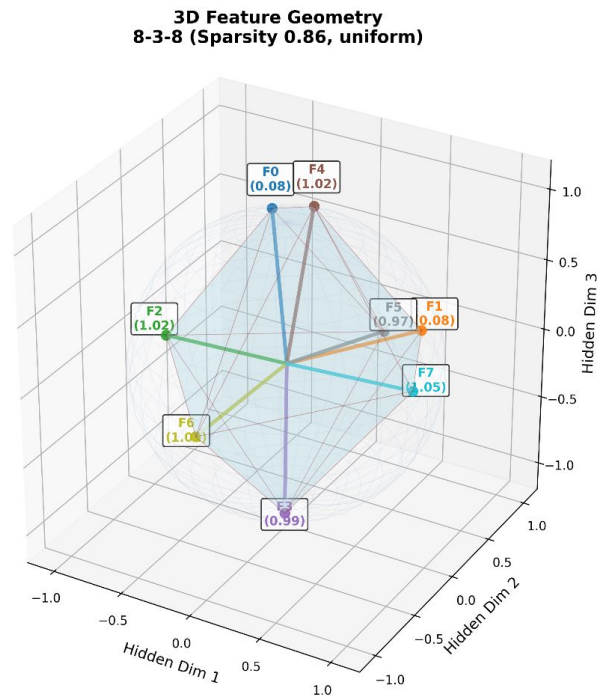
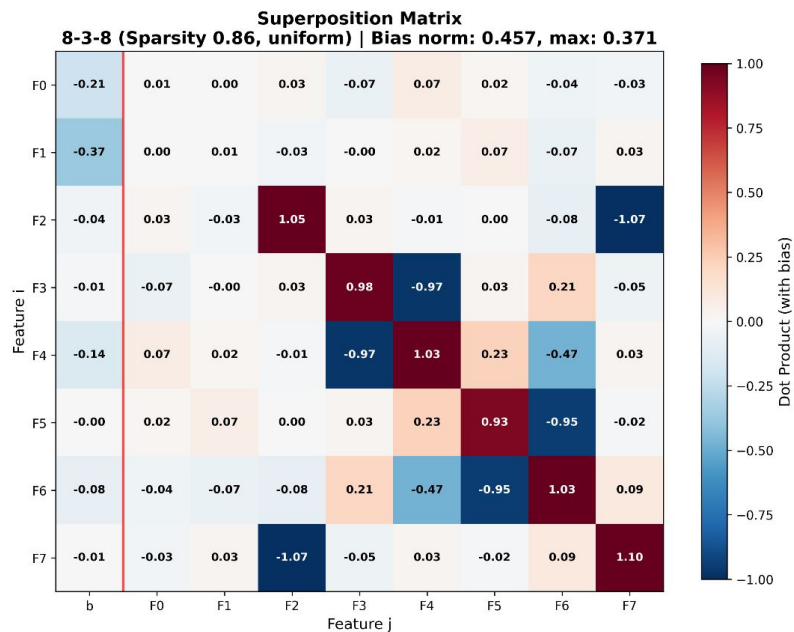
Superposition Matrix $W^T @ W$
5-2-5 (Sparsity 0.83, uniform) | Bias norm: 0.438, max: 0.28

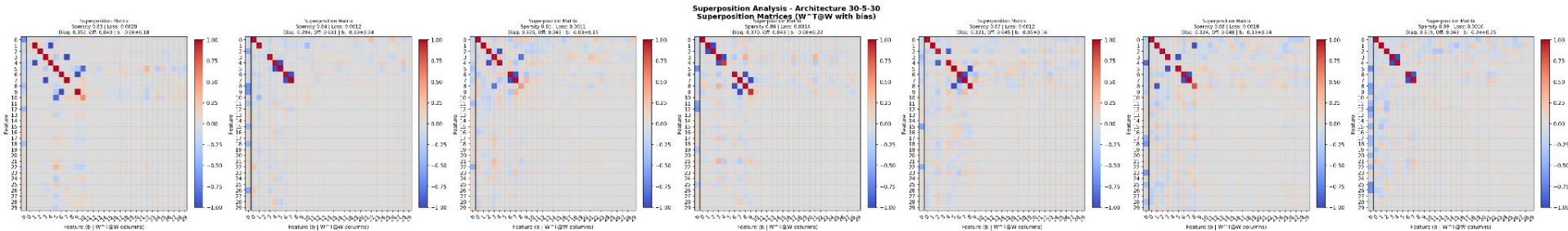
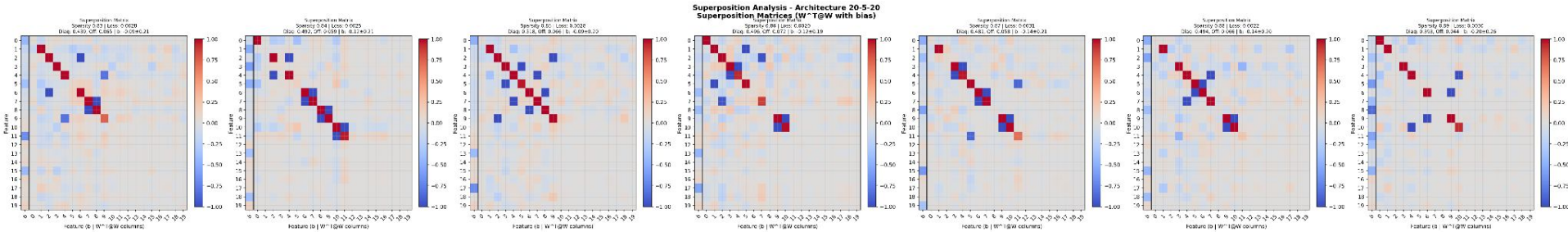
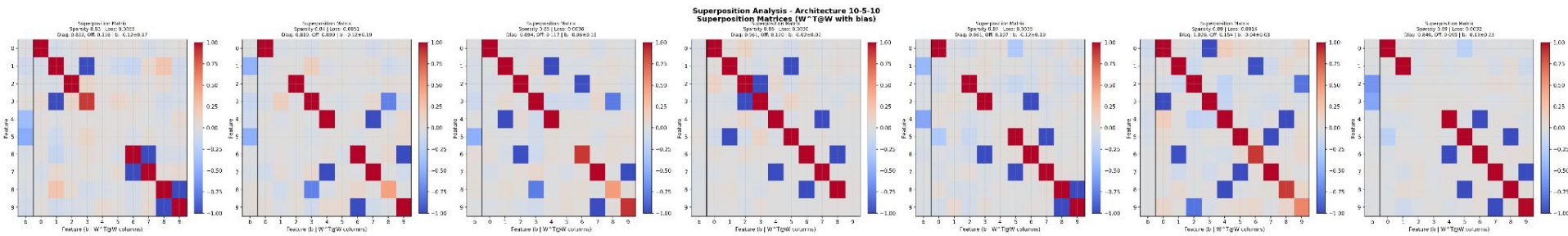


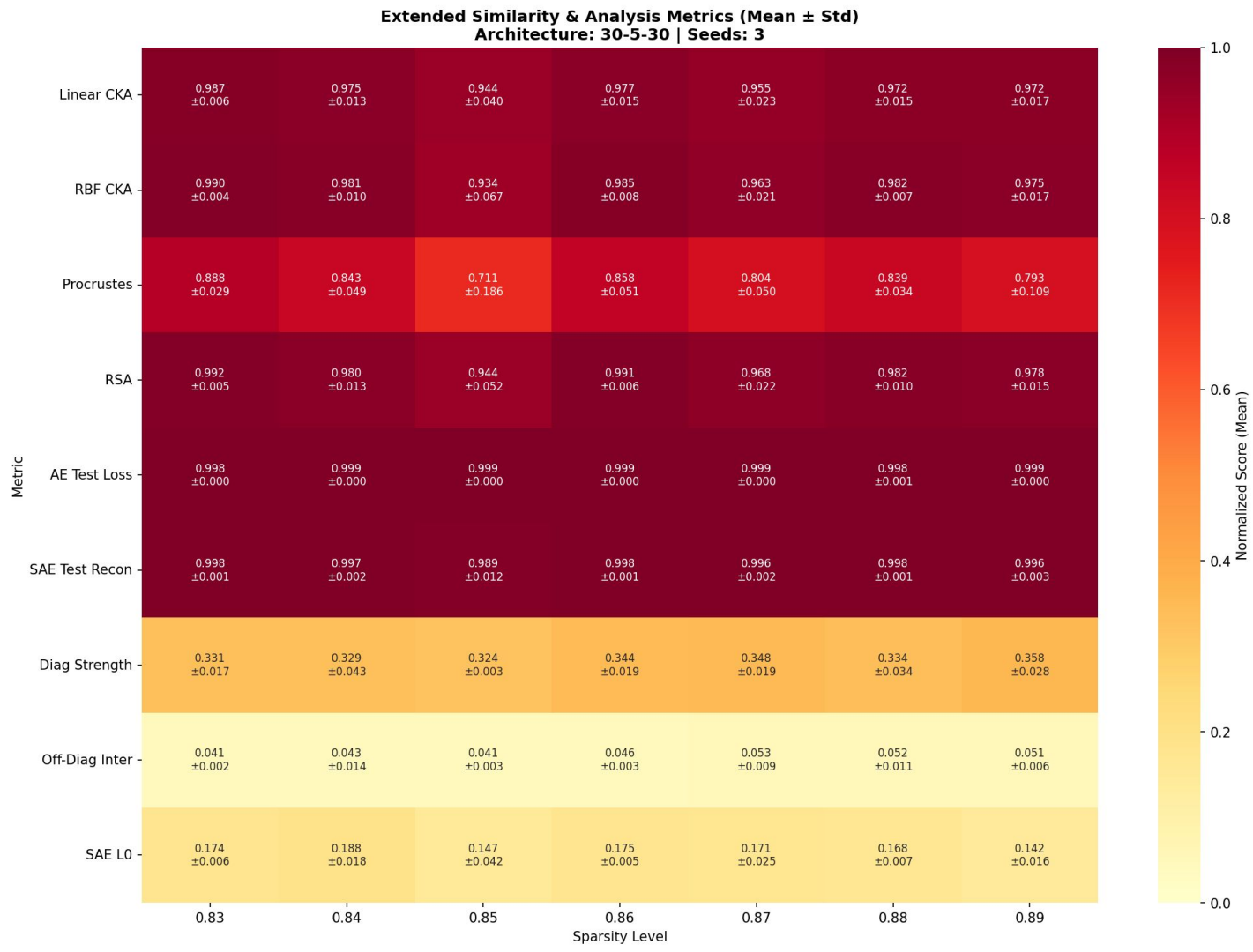
Feature Geometry in 2D Hidden Space

5-2-5 (Sparsity 0.83, uniform)

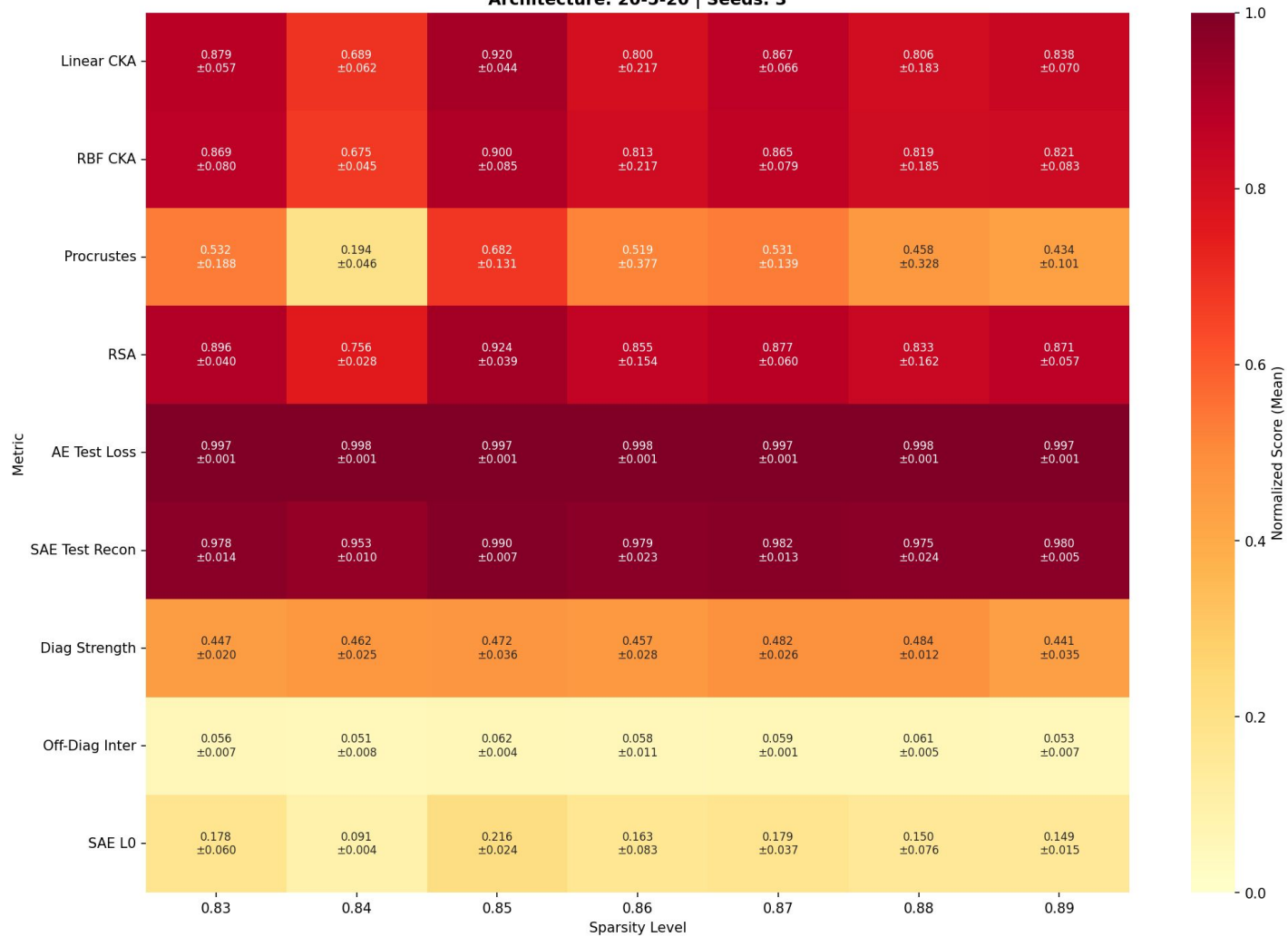




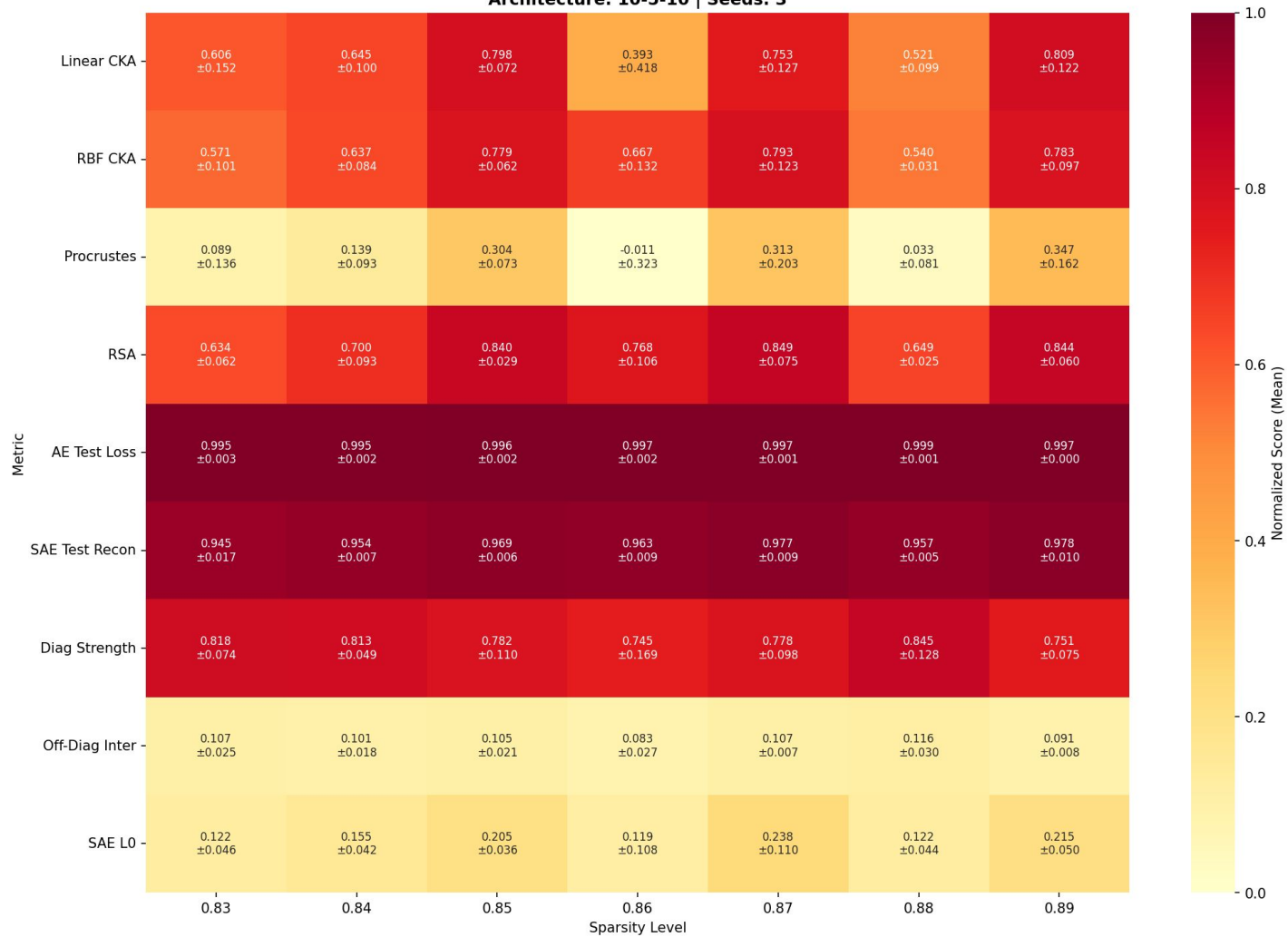




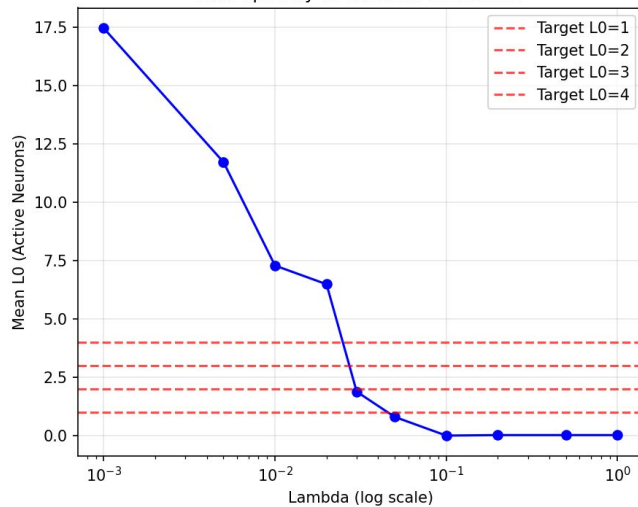
Extended Similarity & Analysis Metrics (Mean $\pm$ Std)
Architecture: 20-5-20 | Seeds: 3



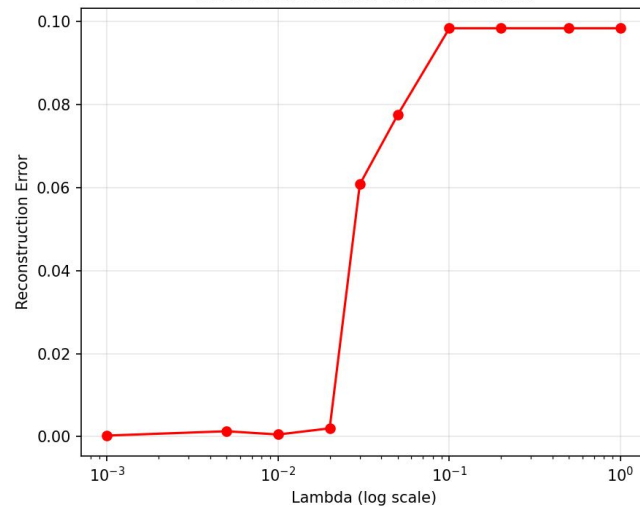
Extended Similarity & Analysis Metrics (Mean $\pm$ Std)
Architecture: 10-5-10 | Seeds: 3



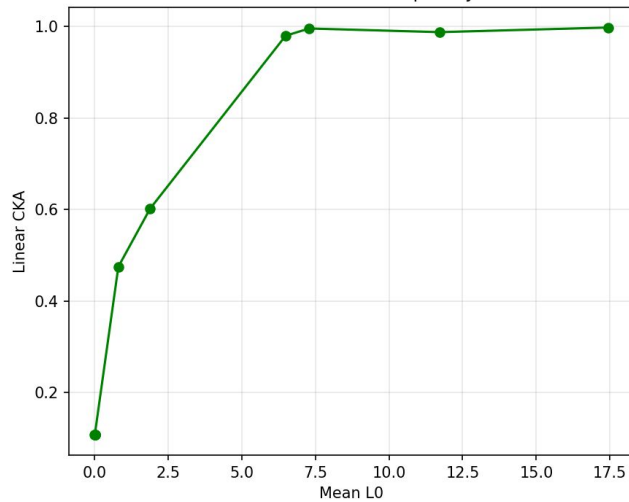
SAE Sparsity vs Lambda Parameter



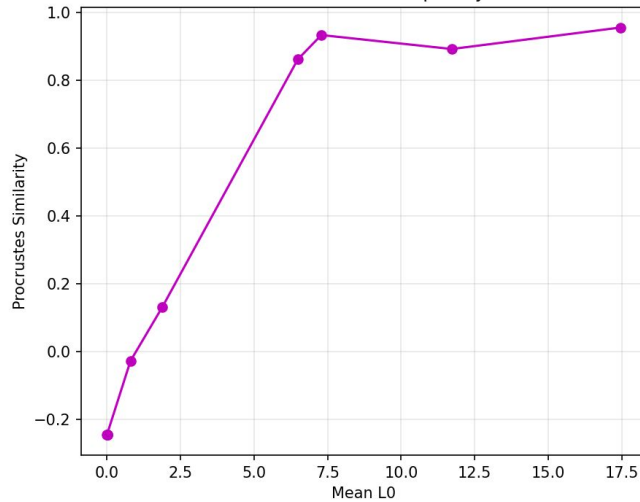
SAE Reconstruction Error vs Lambda



Linear CKA vs SAE Sparsity



Procrustes vs SAE Sparsity



Conclusions

- Phase change
- Feature dictionary (more reliable as dimension increases)
- Irregular pattern
- Procrustes

Future work

- Parameter representation relationship
- Cyclical pattern
- Importances
- Alternative alignment metrics
- Scaling to larger models

References

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<https://arxiv.org/abs/1610.01644>

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https://adamkarvonen.github.io/machine_learning/2024/06/11/sae-intuitions.html

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